

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0186

Sowego Dp 138 kV
February 25, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0186. The Interconnection Customer proposes a 555.7 MW natural gas combined cycle facility interconnecting at the Sowego Dp 138 kV substation in the Virginia area. The request was received on December 19, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 112 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$32,008,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: To Vaca / Thomaston 138kV
Contingency: Blaine / Chaffee Gas Recovery Substation 138kV
Loading: 140.0% (Rating: 159 MVA)
DFAX: 15.8%
Pre-project loading: 84.5%
2. Facility: Woods Dp / Toa Vaca 138kV Line 2
Contingency: Woods Dp / Toa Vaca Substation 138kV Line 1
Loading: 123.1% (Rating: 331 MVA)
DFAX: 18.4%
Pre-project loading: 92.2%
3. Facility: Woods Dp / Vmea Peaking Gen 138kV
Contingency: West Staunton / Woods Dp 138kV Line
Loading: 119.8% (Rating: 234 MVA)
DFAX: 12.6%
Pre-project loading: 89.8%
4. Facility: West Staunton / WoodsDp Substation 138kV
Contingency: Vmea Peaking Gen / Woods Dp 138kV Line
Loading: 119.4% (Rating: 324 MVA)
DFAX: 17.5%
Pre-project loading: 89.4%
5. Facility: Colonial / Idaho Springs 138kV L
Contingency: Colonial / Sideburn 138kV
Loading: 116.1% (Rating: 302 MVA)
DFAX: 12.7%
Pre-project loading: 92.7%

6. Facility: Sideburn / Colonial 138kV
Contingency: Colonial / Idaho Springs 138kV
Loading: 115.8% (Rating: 265 MVA)
DFAX: 9.9%
Pre-project loading: 95.0%
7. Facility: Prince William Dp / N Bluff 138kV
Contingency: N Bluff / Prince William Dp 138kV
Loading: 114.5% (Rating: 260 MVA)
DFAX: 14.4%
Pre-project loading: 83.8%
8. Facility: N Bluff Substation / Prince William Dp 138kV Cir. 2
Contingency: Prince William Dp / N Bluff 138kV #1
Loading: 114.5% (Rating: 260 MVA)
DFAX: 14.4%
Pre-project loading: 83.8%
9. Facility: Prince William Dp / W Staunton 138kV
Contingency: Clover Hill / Elk Horn Substation 138kV
Loading: 112.9% (Rating: 298 MVA)
DFAX: 16.8%
Pre-project loading: 81.4%
10. Facility: Pasco Cnty Solid Waste Resource Recovery / Novec Paradise 138kV
Contingency: West Staunton / Prince William Dp 138kV
Loading: 111.9% (Rating: 240 MVA)
DFAX: 8.5%
Pre-project loading: 92.3%
11. Facility: Woods Dp / Toa Vaca 138kV Cir. 1
Contingency: Woods Dp / Toa Vaca 138kV Circuit 2
Loading: 109.5% (Rating: 372 MVA)
DFAX: 18.4%
Pre-project loading: 82.0%
12. Facility: Jones Chapel 500/230 kV Auto
Contingency: Dekalb / Jones Chapel Substation 500kV
Loading: 108.3% (Rating: 1,091 MVA)
DFAX: 5.5%
Pre-project loading: 105.4%
13. Facility: Norfolk / Battery Heights Dp Substation 138kV Line 1
Contingency: Battery Heights Dp / Norfolk 138kV Cir. 2
Loading: 103.7% (Rating: 333 MVA)
DFAX: 7.3%
Pre-project loading: 91.5%
14. Facility: Norfolk / Clifton 138kV line
Contingency: Clifton / Norfolk 138kV #1
Loading: 102.9% (Rating: 401 MVA)
DFAX: 12.8%
Pre-project loading: 85.1%
15. Facility: Prince William Dp Substation / Clover Hill 138kV L
Contingency: West Staunton / Woods Dp 138kV
Loading: 102.8% (Rating: 232 MVA)
DFAX: 9.4%
Pre-project loading: 80.2%
16. Facility: El Horn / Clover Hill 138kV line
Contingency: West Staunton / Woods Dp Substation 138kV

Loading: 102.8% (Rating: 232 MVA)

DFAX: 9.4%

Pre-project loading: 80.2%

17. Facility: Pleasant Vly Substation / Fountain 138kV

Contingency: Yavapai / Luna 138kV

Loading: 102.4% (Rating: 266 MVA)

DFAX: 9.1%

Pre-project loading: 83.4%

18. Facility: Blaine / Stonewll Energy Ctr Substation 138kV

Contingency: Fountain / Thomaston 138kV

Loading: 102.1% (Rating: 230 MVA)

DFAX: 6.5%

Pre-project loading: 86.3%

19. Facility: Norfolk / Battery Heights Dp 138kV Line 2

Contingency: Norfrk / Battery Hts Dp 138kV

Loading: 101.7% (Rating: 339 MVA)

DFAX: 7.3%

Pre-project loading: 89.8%

20. Facility: Blaine / ChaffeeGas Recovery 138kV

Contingency: Fountain / Thomaston 138kV

Loading: 101.5% (Rating: 264 MVA)

DFAX: 6.5%

Pre-project loading: 87.8%

21. Facility: Norfolk / Clifton 138kV Line 1

Contingency: Norfolk / Clifton 138kV Line 2

Loading: 100.3% (Rating: 412 MVA)

DFAX: 12.8%

Pre-project loading: 83.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Thomaston / Toa Vaca 138kV L	Rebuild line with higher-capacity conductor	\$1,809,000	74% (\$1,338,000)
Woods Dp / Toa Vaca 138kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,404,000	74% (\$1,039,000)
Woods Dp / Vmea Peaking Gen 138kV	Rebuild line with higher-capacity conductor	\$5,891,000	80% (\$4,691,000)
West Staunton / Woods Dp 138kV	Rebuild line with higher-capacity conductor	\$884,000	74% (\$654,000)
Idaho Springs / Colonial 138kV	Rebuild line with higher-capacity conductor	\$1,914,000	67% (\$1,286,000)
Sieburn / Colonial 138kV line	Rebuild line with higher-capacity conductor	\$4,185,000	80% (\$3,364,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Prince William Dp / N Bluff 138kV	Rebuild line with higher-capacity conductor	\$3,386,000	79% (\$2,684,000)
N Bluff Substation / Prince William Dp 138kV Cir. 2	Rebuild line with higher-capacity conductor	\$6,707,000	79% (\$5,316,000)
W Staunton / Prince William Dp 138kV	Rebuild line with higher-capacity conductor	\$1,012,000	79% (\$801,000)
Pasco Cnty Solid Waste Resource Recovery / Novec Paradise 138kV	Rebuild line with higher-capacity conductor	\$4,890,000	67% (\$3,294,000)
Woods Dp / Toa Vaca 138kV Line 1	Reconductor line	\$501,000	74% (\$371,000)
Jones Chapel 500 kV/230 kV Autotransformer	Replace transformer with higher-rated unit	\$20,803,000	9% (\$1,856,000)
Norfolk / Battery Heights Dp Substation 138kV Line 1	Reconductor line	\$278,000	81% (\$225,000)
Norfolk / Clifton 138kV line	Reconductor line	\$674,000	74% (\$497,000)
Prince William Dp Substation / Clover Hill 138kV L	Reconductor line	\$666,000	80% (\$531,000)
Clover Hill / Elk Horn 138kV	Reconductor line	\$560,000	80% (\$447,000)
Pleasant Vly Substation / Fountain 138kV	Reconductor line	\$1,937,000	81% (\$1,565,000)
Blaine / Stonewall Energy Center 138kV	Reconductor line	\$556,000	82% (\$453,000)
Battery Heights Dp Substation / Norfrk 138kV L	Reconductor line	\$302,000	81% (\$245,000)
Blaine / Chaffee Gas Recovery 138kV	Reconductor line	\$922,000	82% (\$751,000)
Norfolk / Clifton 138kV Line 1	Reconductor line	\$814,000	74% (\$600,000)
	Total allocated to project		\$32,008,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.